

Temporal Entropy Analysis of Traffic Flow Dynamics

Complex Systems ePortfolio

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1 Introduction

Entropy-based methods are commonly used to study complexity, uncertainty, and predictability in dynamical systems. Many real-world systems, including traffic systems, exhibit changing behaviour over time due to interactions between multiple underlying processes. Traffic flow is influenced by factors such as commuting behaviour, congestion buildup, road capacity, accidents, and temporal activity patterns. As a result, traffic dynamics are not perfectly regular and may fluctuate substantially across time.

Traditional traffic analysis often focuses on average traffic volume or peak traffic periods. However, these approaches may overlook changes in temporal organization and predictability within the signal itself. Entropy-based methods provide an alternative approach by quantifying how regular, uncertain, or complex the observed dynamics are over time. Entropy measures are useful for traffic analysis because traffic systems alternate between more stable and more irregular regimes across time.

This study examines temporal changes in traffic entropy using Shannon entropy and Sample Entropy.

Research Question: Are there changes over time in the entropy estimates, and what might these changes indicate?

Shannon entropy originates from information theory and measures uncertainty in the probability distribution of observations (Shannon, 1948). Higher Shannon entropy values indicate greater uncertainty and variability within the observed traffic states, while lower values suggest more repetitive or concentrated behaviour. In contrast, Sample Entropy measures temporal regularity within a time series rather than only distributional uncertainty. Lower Sample Entropy values indicate stronger regularity and predictability in the temporal

dynamics, whereas higher values indicate more irregular behaviour (Richman and Moorman, 2000).

Recent methodological work has further demonstrated that entropy-based measures are useful tools for analysing complexity and temporal organization in nonlinear systems (Delgado-Bonal and Marshak, 2019). Entropy measures can reveal changes in predictability and dynamical regularity that may not be visible through standard statistical summaries alone. This makes entropy-based analysis especially suitable for studying evolving real-world systems such as traffic flow, where system behaviour may shift between more stable and more irregular regimes over time.

Together, Shannon entropy and Sample Entropy provide complementary perspectives on traffic dynamics. Shannon entropy captures uncertainty in the distribution of observations, whereas Sample Entropy captures regularity in temporal patterns and sequential behaviour.

2 Description of the dataset

The dataset consists of traffic volume measurements collected during March 2026 at 15-minute intervals. Each day contains 96 observations corresponding to 15-minute aggregation windows across a 24-hour period. Daily files were combined into continuous time series from multiple traffic detector locations.

The dataset is temporally ordered, continuous, and evenly sampled, making it suitable for entropy-based time series analysis. Because the observations span an entire month, the dataset contains repeated daily cycles as well as variations across different days and traffic conditions.

Traffic systems are inherently dynamic because traffic flow changes continuously across time. During some periods, traffic behaviour may be relatively stable and predictable, while during other periods the system may become more variable and irregular. These changing conditions make traffic flow an appropriate system for studying temporal fluctuations in entropy and complexity.

Although the analysis only includes a limited number of detector locations, the dataset still captures important temporal variations in traffic organization over time. The continuous structure of the dataset allows entropy measures to be applied across moving windows in order to investigate how predictability and temporal regularity evolve throughout the observation period.

3 Methodology

The traffic volume signal was analysed using Shannon entropy and Sample Entropy. Both methods were applied across temporal windows to examine how entropy changes over time rather than assuming stationary dynamics.

Before analysis, the daily observations were combined into continuous time series. The signals were inspected for missing values and potential irregularities prior to analysis in order to reduce distortions in the entropy estimates caused by preprocessing issues. The analyses were performed on multiple traffic sites in order to compare temporal entropy dynamics across locations.

Shannon entropy was calculated across temporal windows to estimate uncertainty in traffic states. The method quantifies how evenly distributed the observations are within a given window. Higher Shannon entropy values indicate more diverse and uncertain traffic conditions, while lower values indicate more repetitive and concentrated traffic states.

Shannon entropy is useful because it provides a relatively simple measure of uncertainty within a system. However, it does not directly incorporate temporal ordering. Two signals may exhibit similar distributions while having different temporal structures. For this reason, Shannon entropy was complemented with Sample Entropy analysis.

Sample Entropy was computed using a sliding-window approach to measure temporal regularity within the signal. Entropy values were calculated across overlapping temporal windows in order to capture gradual changes in traffic dynamics through time. A rolling window size of 96 observations was used, corresponding to one full day of traffic activity. The method evaluates the probability that similar sequences of observations remain similar at the next time step. Lower Sample Entropy values indicate stronger temporal organization and predictability, while higher values indicate more irregular dynamics (Richman and Moorman, 2000).

Sample Entropy was calculated using embedding dimension $m = 2$ and tolerance parameter $r = 0.2$ times the standard deviation of the signal. These parameter choices are commonly used in entropy-based time series analysis (Delgado-Bonal and Marshak, 2019).

Unlike Shannon entropy, Sample Entropy explicitly incorporates sequential structure and therefore provides information about temporal organization within the traffic signal. This makes it particularly useful for identifying periods where traffic dynamics become either highly regular or highly irregular.

Entropy-based approaches have also been applied specifically to traffic systems in order to analyse changing traffic complexity and temporal dynamics (Shang et al., 2017). Previous research has shown that entropy measures can capture fluctuations in traffic regularity and help characterize transitions between different traffic regimes. This provides additional motivation for applying entropy-based methods to traffic flow analysis in the present study.

The entropy measures were computed using sliding temporal windows across the traffic signal. This approach allows the investigation of evolving system behaviour rather than only calculating a single global entropy estimate for the entire dataset.

Because entropy estimates can depend on methodological choices such as window size and parameter selection, the present analysis should be interpreted primarily as an exploratory investigation of temporal organization within traffic dynamics rather than a complete characterization of the underlying system.

4 Results

Both entropy measures varied across time, suggesting that traffic dynamics are non-stationary rather than fully stable over the observation period.

Rolling Shannon entropy reflected changes in the diversity and uncertainty of traffic states across different locations. Most sites exhibited relatively stable entropy values with moderate fluctuations through time. However, several temporary decreases in Shannon entropy were observed, particularly for Site 164, indicating periods where the distribution of traffic states became more concentrated and predictable.

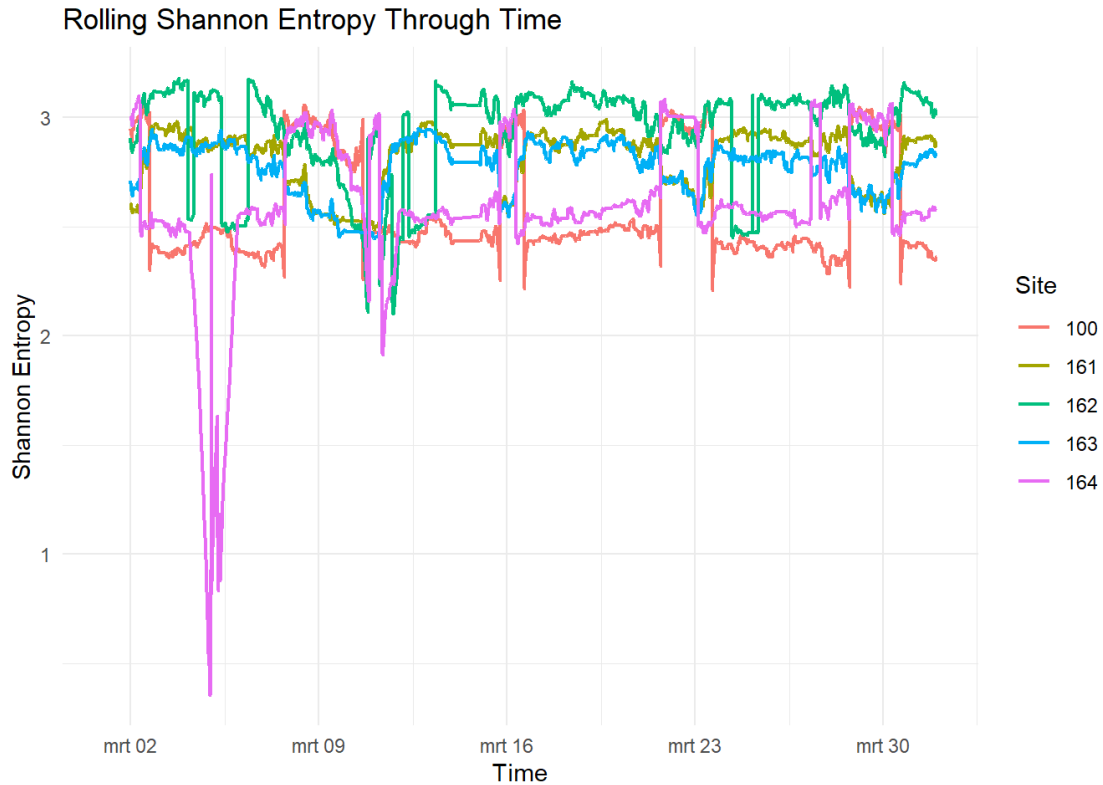


Figure 1: Rolling Shannon entropy estimates through time for multiple traffic sites.

Rolling Sample Entropy revealed clearer temporal variation in signal regularity and predictability. Lower Sample Entropy values indicate more regular temporal dynamics, while higher values indicate greater irregularity. A substantial temporary decrease in Sample Entropy was observed around March 4 for Site 164, where entropy approached zero, suggesting a short period of highly regular and repetitive traffic behaviour.

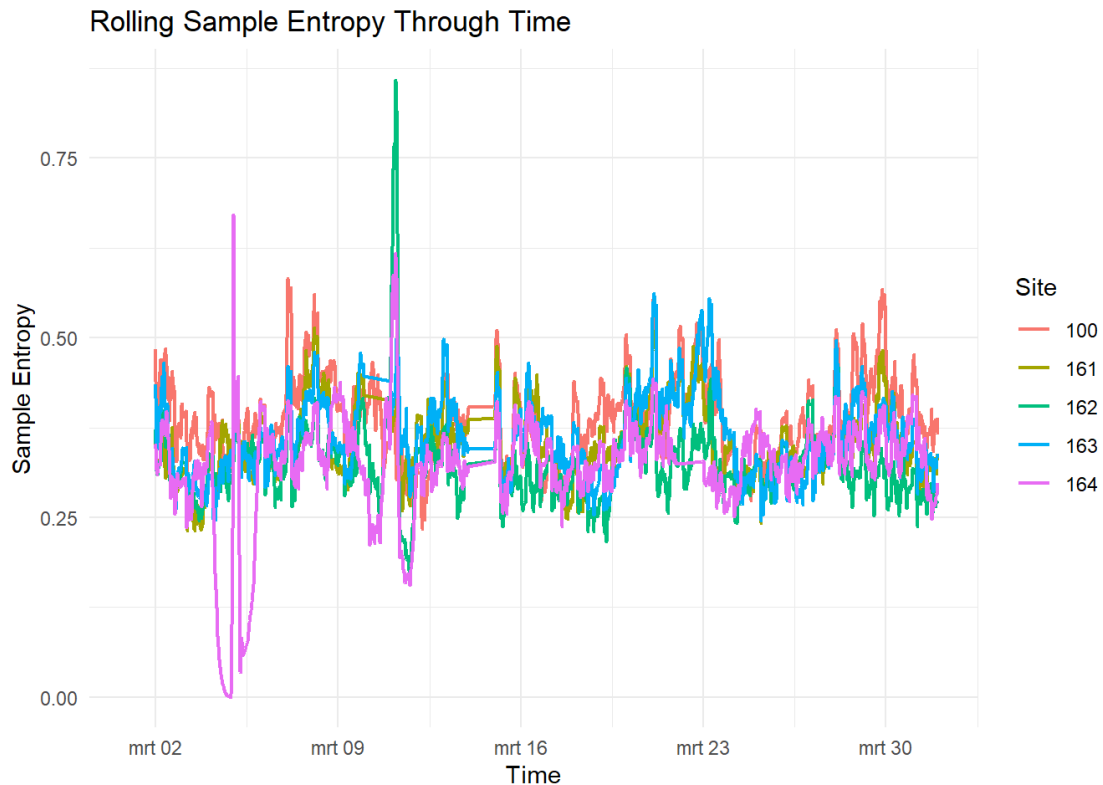


Figure 2: Rolling Sample Entropy estimates through time for multiple traffic sites.

Although both entropy measures capture aspects of uncertainty and complexity, they describe different properties of the traffic system. Shannon entropy measures the diversity of traffic states within a temporal window, whereas Sample Entropy measures temporal regularity and predictability in the evolution of the signal itself.

5 Discussion

The results suggest that traffic dynamics evolve over time with changing levels of predictability and temporal structure. Entropy-based methods therefore provide insight into evolving system behaviour beyond simple average traffic measurements.

The observed fluctuations in entropy may reflect changes in traffic organization caused by varying human activity patterns, congestion dynamics, or other external influences. Lower

Sample Entropy values corresponded to periods of stronger regularity and predictability, while higher values reflected more irregular and less predictable temporal dynamics.

The sharp temporary decrease in Sample Entropy observed for Site 164 may indicate a short interval of unusually regular traffic dynamics. One possible explanation is that traffic flow during this period became highly repetitive due to stable commuting patterns or reduced variability in traffic conditions. Because entropy measures are sensitive to temporal regularity, highly repetitive sequences can produce very low Sample Entropy values.

An important advantage of entropy-based analysis is that it does not require explicit assumptions about the exact mechanisms generating the traffic signal. Instead, entropy measures provide a general framework for quantifying uncertainty and temporal organization within complex systems.

At the same time, the analysis remains exploratory and depends strongly on methodological choices such as window size, entropy parameters, and preprocessing decisions. Different parameter selections may produce somewhat different entropy estimates and interpretations.

Another limitation is that the study used data from only a limited number of traffic locations. Traffic systems are spatially distributed systems influenced by interactions between multiple locations and external conditions. Future work could therefore compare entropy dynamics across larger traffic networks or additional detector sites.

Additional nonlinear methods could also complement entropy analysis. For example, recurrence analysis or phase space reconstruction could provide further insight into the underlying temporal organization of traffic dynamics.

6 Conclusion

This study applied Shannon entropy and Sample Entropy to analyse temporal changes in traffic flow dynamics. The results showed that entropy estimates vary across time, indicating changing levels of uncertainty and temporal organization within the traffic system.

Shannon entropy quantified changes in uncertainty and diversity within traffic states, while Sample Entropy captured changes in temporal regularity and predictability. The analyses revealed periods of both relatively stable and more irregular traffic behaviour across the observation period.

In particular, the temporary decrease in Sample Entropy around March 4 suggested a short period of increased regularity and predictability within the traffic dynamics. More broadly, the results demonstrated that traffic systems exhibit evolving temporal structure rather than purely stationary behaviour.

Overall, the findings suggest that entropy-based methods can detect temporal changes in predictability and organization within traffic systems and provide useful insight into evolving

dynamics in complex real-world systems such as traffic flow. These findings illustrate how traffic systems can exhibit changing levels of order, predictability, and temporal organization characteristic of complex dynamical systems.

AI Usage Statement

Generative AI tools were used for language editing and structuring the text. The analysis, interpretation of results, and methodological decisions were reviewed and developed by the authors.

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